

19.6 Collinearity, previewed

Collinearity, near-linear dependence among the columns of $\mathbf{X}$, violates none of the assumptions of Part III. As long as $\mathbf{X}$ has full rank, every result of Parts II and III holds. What collinearity does is make some of them unhelpful: intervals that are correct but very wide. Chapter 26 develops diagnostics, and Chapter 27 remedies that trade bias for variance.

19.6.1 Variance inflation

For a single coefficient, the Frisch–Waugh–Lovell theorem gave Eq. 6.6.2:

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{S_{jj}} \cdot \frac{1}{1 - R_j^2},$$

where S_{jj} is the centred sum of squares of $\mathbf{x}_j$ and R_j^2 is the coefficient of determination from regressing $\mathbf{x}_j$ on the other regressors. The first factor is the variance $\mathbf{x}_j$ would give if it were orthogonal to the other regressors. The second, the **variance inflation factor** $\text{VIF}_j = 1/(1 - R_j^2)$, is the price of the part of $\mathbf{x}_j$ that the others can already explain. A variance inflation factor of 100 means the standard error is ten times what an orthogonal design with the same spread would give.

The factors have a compact matrix form. Let $\mathbf{R}$ be the correlation matrix of the k regressors other than the intercept. Then $\text{VIF}_j = [\mathbf{R}^{-1}]_{jj}$ (Exercise (A1)), and the eigenvalues of $\mathbf{R}$ show where the problem lies.

Proposition 19.6.1 — Where collinearity costs variance

Let $\mathbf{X}$ have full rank and let $\mathbf{X}^\top \mathbf{X} = \sum_{\ell=1}^p \lambda_\ell \mathbf{v}_\ell \mathbf{v}_\ell^\top$ with orthonormal eigenvectors $\mathbf{v}_\ell$ and eigenvalues $\lambda_\ell > 0$.

- (a) For every $\mathbf{a} \in \mathbb{R}^p$, $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \sum_{\ell} (\mathbf{a}^\top \mathbf{v}_\ell)^2 / \lambda_\ell$.
- (b) The fitted values at the design points have average variance $\sigma^2 p/n$, whatever the eigenvalues.
- (c) With $\mathbf{R}$ as above, $\sum_j \text{VIF}_j = \text{tr } \mathbf{R}^{-1} = \sum_{\ell} 1/\lambda_\ell(\mathbf{R})$, and every $\text{VIF}_j \leq 1/\lambda_{\min}(\mathbf{R})$.

Proof.

- (a) By Theorem 1.7.2, $(\mathbf{X}^\top \mathbf{X})^{-1} = \sum_{\ell} \lambda_\ell^{-1} \mathbf{v}_\ell \mathbf{v}_\ell^\top$, and $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}$ (Corollary 5.5.2). (b) $\text{Cov}(\hat{\mathbf{Y}}) = \sigma^2 \mathbf{M}$, and $\text{tr } \mathbf{M} = p$ (Proposition 6.3.4). (c) The first equality is $\text{VIF}_j = [\mathbf{R}^{-1}]_{jj}$ summed over j . The trace is the sum of the eigenvalues of $\mathbf{R}^{-1}$, which are $1/\lambda_\ell(\mathbf{R})$. Each diagonal entry of $\mathbf{R}^{-1}$ is at most its largest eigenvalue, by Theorem 1.7.7 applied to a standard basis vector.

■

Part (a) locates the damage. A small eigenvalue inflates the variance of $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$ only through the component of $\mathbf{a}$ along its eigenvector. Individual coefficients usually have such components, and so do predictions at points that break the pattern of the regressors; predictions that follow the

pattern do not, and by (b) fitted values at the design points are as precise on average as in any design. Collinearity is a problem about *attributing* an effect to one of several regressors that moved together, not about fitting.

Example. (Longley's regression)

Longley's (1967) data, which Example (Longley's data, exactly) used to test the accuracy of least squares algorithms, regress total employment on six economic series over $n = 16$ years. The variance inflation factors range from 3.59 for ARMED (the size of the armed forces) to 1788.5 for GNP. The eigenvalues of the correlation matrix run from 4.603 down to 0.00038, and the factors sum to 3119.4, which is the sum of the reciprocal eigenvalues, as [Proposition 19.6.1\(c\)](#) requires. Four of the series (the price deflator, GNP, population and the year itself) rose almost in step over the sixteen years, each with correlation above 0.99 with the year. The data can say much about their joint movement and almost nothing about any one of them with the others held fixed.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data
Z = data.drop(columns="TOTEMP").to_numpy()          # six regressors
n, k = Z.shape

def r_squared(v, W):
    W1 = np.column_stack([np.ones(len(v)), W])
    fitted = W1 @ np.linalg.lstsq(W1, v, rcond=None)[0]
    return 1 - np.sum((v - fitted) ** 2) / np.sum((v - v.mean()) ** 2)

vif = np.array([1 / (1 - r_squared(Z[:, j], np.delete(Z, j, axis=1))) for j in range
(k)]]
R = np.corrcoef(Z, rowvar=False)
print(dict(zip(data.columns[1:], np.round(vif, 1))))
print("diagonal of R^{-1}:", np.round(np.diag(np.linalg.inv(R)), 1))
print("eigenvalues of R: ", np.round(np.linalg.eigvalsh(R), 5))
```

[Run this code online](#)

Example. (Predicting along and across the pattern)

Draw $m = 40$ pairs of regressors with sample correlation 0.9868, so that each has a variance inflation factor of 38.1, and fit $E(Y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$. In units of σ , the two slopes have standard deviations 0.904 and 0.907. At the new point $(x_1, x_2) = (1, 1)$, which follows the pattern of the data, the fitted mean has standard deviation 0.228, close to the root mean square over the design points, $\sqrt{3/40} = 0.274$. At $(1, -1)$, which is just as far from the origin but against the pattern, it is 1.809. [Figure 19.6.1](#) shows the contours of this standard deviation. They are long ellipses aligned with the data cloud.

```
rng = np.random.default_rng(1906)
m, rho = 40, 0.98
x1 = rng.normal(size=m)
x2 = rho * x1 + np.sqrt(1 - rho**2) * rng.normal(size=m)
X = np.column_stack([np.ones(m), x1, x2])
G = np.linalg.inv(X.T @ X)          # Cov(beta_hat) / sigma^2
along = np.array([1.0, 1.0, 1.0])  # a new point that follows the
```

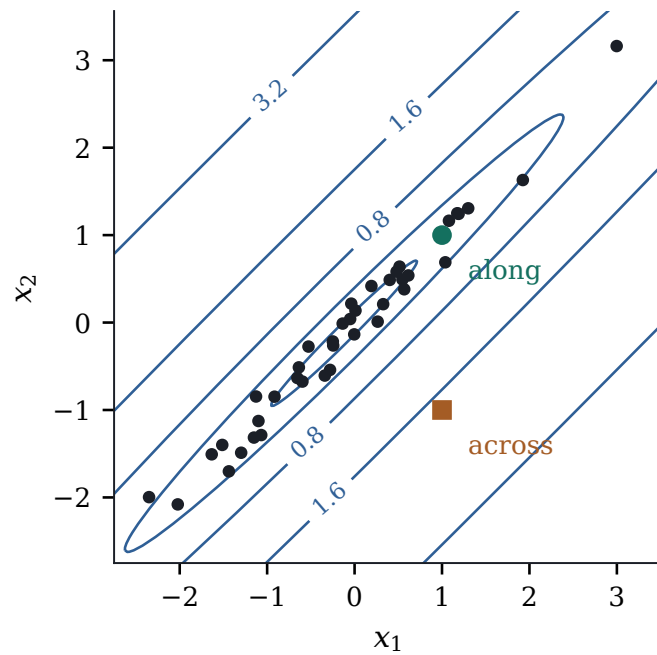


Figure 19.6.1: Standard deviation of the fitted mean, in units of σ , over the regressor plane for [Example \(Predicting along and across the pattern\)](#), with the 40 design points. The contours are elongated along the data: prediction at the circle, which follows the pattern, is precise; prediction at the square, which breaks it, is not.

```

pattern
across = np.array([1.0, 1.0, -1.0]) # same distance, against the
pattern
print("sd of beta_1, beta_2 (units of sigma):", np.round(np.sqrt(np.diag(G)[1:]), 3)
)
print("sd of the fitted mean, along :", round(np.sqrt(along @ G @ along), 3))
print("sd of the fitted mean, across:", round(np.sqrt(across @ G @ across), 3))
print("average over the design points:", round(np.trace(X @ G @ X.T) / m, 3), "= p/n
")

```

[Run this code online](#)

19.6.2 How collinearity interacts with the other departures

Collinearity causes no bias, but it amplifies the departures of the earlier sections. By Eq. 19.1.2 the stakes of omitting x_2 are $r^2|\gamma - 1|$, so the choice between short and long models matters most exactly when the data are least able to make it. Perturbations of the regressors, from rounding or measurement error, are governed by the condition number of $\mathbf{X}$ (Theorem 10.5.1), which is large for the same reason. And a single case can have high leverage without being extreme in any one regressor, because leverage measures distance in the metric of the regressors' covariance (Proposition 6.8.2).

Idea. Collinearity is a shortage of information, not a failure of the model, and it magnifies every other departure.

19.6.3 Exercises

19.6.3.1 A. Check your understanding

Exercise A1.

Let $\mathbf{Z}$ be the $n \times k$ matrix of centred regressors, each scaled to unit length, so that $\mathbf{Z}^\top \mathbf{Z} = \mathbf{R}$. Use Eq. 6.6.1 to show that $[\mathbf{R}^{-1}]_{jj} = 1/(1 - R_j^2)$.

Solution. By Eq. 6.6.1 with $\mathbf{X}_1 = \mathbf{z}_j$ and $\mathbf{X}_2$ the other columns, $[\mathbf{R}^{-1}]_{jj} = 1/\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{z}_j\|^2$, where $\mathbf{M}_{(j)}$ projects onto the other centred columns. Since $\mathbf{z}_j$ is centred with unit length, the squared residual length is $1 - R_j^2$, as in Exercise (B1) (the intercept only removes means, and the columns are already centred).

Exercise A2.

Show that $\text{VIF}_j \geq 1$, with equality iff the centred $\mathbf{x}_j$ is orthogonal to every other centred regressor.

19.6.3.2 B. Practice

Exercise B1.

For two centred regressors with $\|\mathbf{x}_1\|^2 = \|\mathbf{x}_2\|^2 = S$ and correlation r , show that $\text{Var}(\hat{\beta}_1 + \hat{\beta}_2) = 2\sigma^2/(S(1+r))$ and $\text{Var}(\hat{\beta}_1 - \hat{\beta}_2) = 2\sigma^2/(S(1-r))$. Relate the result to [Proposition 19.6.1\(a\)](#) and to [Figure 19.6.1](#).

Solution. The slope block of $\mathbf{X}^\top \mathbf{X}$ is $S \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}$, with eigenvalues $S(1+r)$ and $S(1-r)$ and eigenvectors $(1, 1)^\top/\sqrt{2}$ and $(1, -1)^\top/\sqrt{2}$. The intercept is orthogonal to the centred slopes and plays no role. With $\mathbf{a} = (1, \pm 1)^\top$, $\mathbf{a}^\top \mathbf{v}$ is $\sqrt{2}$ on one eigenvector and 0 on the other, so part (a) gives $2\sigma^2/(S(1 \pm r))$. When r is near one, the sum, the combined effect of two regressors that move together, is well determined, and the difference is not. In the figure, the new point $(1, 1)$ asks for the sum and $(1, -1)$ for the difference.

Exercise B2.

Show that $\sum_j \text{VIF}_j \geq k$, and that $\max_j \text{VIF}_j \geq (\sum_j \text{VIF}_j)/k$. Deduce that a correlation matrix with a small eigenvalue must have at least one large variance inflation factor, and conversely.

19.6.3.3 C. Going deeper

Exercise C1.

Show that if $\mathbf{x}_0$ is a convex combination of the rows of $\mathbf{X}$, then $\text{Var}(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}) \leq \sigma^2 \max_i h_{ii}$. Is the converse true: does a small prediction variance mean $\mathbf{x}_0$ lies inside the data?

Compare with Example (Hidden extrapolation).